CS6750: Homework 1

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1 QUESTION 1

In this question, the goal is to follow along with a cooking video recipe. Video streaming helps ensure you know the steps in order, but most of the time, the video moves faster than the participants executing the recipe at home.

1.1 Preparing Ingredients

At first glance, this context feels pretty forgiving. Your hands are free, nothing's on the stove yet, so you can tap, pause, and rewind the video as much as you want. But there's still a catch: this stage is where you're translating a list of ingredients on screen into physical objects in your fridge and pantry, and that mapping isn't always obvious. The video might flash "2 cloves of garlic, minced" for two seconds before moving on, and now you're digging through the fridge trying to remember if it was two cloves or a tablespoon, and whether you're supposed to mince it now or later. Unlike the active cooking stage where the problem is physical (oily hands, divided attention), the problem here is more about working memory. You're holding onto a growing list of things in your head while walking back and forth between your screen and your fridge, and every time you glance away to grab something, you risk forgetting what came right after it.

A good redesign here would lean into the fact that your hands and eyes are actually free at this stage, unlike during cooking. The app could auto-generate a running checklist that stays pinned on screen (pulled straight from the video) so you're not relying on memory alone, and let you check off each ingredient as you grab it.

1.2 Actively Cooking

Imagine this: you get your hands all oily, and suddenly the task gets way harder because you have to keep toggling back and forth between frames while your greasy fingers make the screen less responsive. And don't even get me started on the moments when you have to split your focus. For example, stir frying with one eye while trying to catch the next ingredient or step on the screen with the

other. Or, the need to process the ingredients while seeing how it is chopped. This is a perfect example of where being a participant in the process actually matters more than the senses we're given as humans.

A good enhancement here would be a video streaming app that actually waits for you to reach a certain stage before moving on. In addition to that a voice assistant feature, so you can just ask, "What's next after seasoning this steak?" without missing a beat. This gives the aspect of user participating in the whole process.

2 QUESTION 2

2.1 Wide Gulf of Execution

Let's talk about the Garmin Instinct Solar 2. For context: five buttons, no touch-screen. Yet somehow it packs in all the usual smartwatch features, such as heart rate, exercise tracking, calories, notifications, payments, stopwatch, find my phone. Plenty of functionality, but only five buttons and no screen to tap.

Without a salesperson walking me through it, I wouldn't have figured out half the things this thing can do. The real limitation here is the interaction itself. Clicking a button on your wrist just isn't natural. People have been taking watches off just to adjust the time for decades, thus for someone to navigate through multiple menus by clicking a button near your wrist is not something general population expects. And to make things worse, each button changes role depending on what screen you're on. For example, to actually pay with this watch, I need one long press followed by ten clicks just to get there. Ten clicks for something that's supposed to be quick and convenient. This is a textbook case of high usefulness, low usability. The features are all there, but the interaction design makes them painfully hard to reach.

2.2 Narrow Gulf of Execution

An example of this is rice cooker. This thing is a prime example of usability done right. One button. That's it. Press it and it starts cooking. The instruction is very simple too, there's a water line printed on the inside, so you don't have to guess how much water to pour.

Following is how a generic idea can have multi-purpose. Once you've mastered the essence of the tool, you can use it for all sorts of things, like sous vide, baking

a cake, steaming food, reheating leftovers like a microwave, even fermenting with its built-in humidity. These multi functionalities can be achieved by one button without too many configurations.

Lesson learnt for me, if the interface is limited, the functionality should either be reduced or made generic enough that the user can take the core idea and run with it. Take the smartwatch example, it has a dedicated stopwatch and an interval mode that beeps every minute for exercise. In my opinion, that interval mode is redundant. A stopwatch could easily do the same job. If they'd just stripped that out, that's one less click to get to what you actually want.

3 QUESTION 3

3.1 AI as Therapist

In my opinion, it's perfectly fine to use AI models as a therapist, as long as the user has the right mindset and the platform has proper guardrails in place. Here's why: not every culture sees therapy as a normal thing. Some people feel judged and end up never seeking any mental help at all. Realistically, not everyone has the extra money or time to spend on traditional therapy. The fact that someone is willing to reach out to an AI for this is already a huge step. It means they're acknowledging they need some form of support, and that alone matters.

Guardrails are the critical piece here. Suppose the AI can switch into a therapy mode where it gently steers people toward a more positive headspace, or even nudges them to see a physical therapist, talk to family, or reach out to friends. Any discussion around self-harm or negativity should have safeguards that gradually guide users toward happier or more constructive topics. A disclaimer is a must. The AI should clearly state it's not an expert, it can be wrong, and users should always seek another perspective from friends, family, or community.

Anything good will have a risks, users might draw analogies to their problems that the AI simply can't pick up on as part of a consultation process. But then again, if someone is doing that, they're probably "smart" enough to understand that AI can be wrong and even manipulated. And honestly, that group is likely smaller than the broader audience that genuinely benefits from this. Another risk is that users might ignore the fact that AI-generated responses can be flawed. One safeguard could be the AI gradually responding shorter and shorter until it

refuses to discuss the topic altogether. This way, users get a brief fresh perspective without being exposed to too much risk from AI acting as a therapist.

In my personal job hunting experience, I consulted AI about my next move. AI gave me a different perspective. It challenged me, asked questions I hadn't considered, and helped me think through points I'd missed. Whether I followed its advice or not, I walked away with more data and a better-informed decision. The idea is to give people more information so they can make smarter choices. That's free speech in action. At the end of the day, it comes down to self-discipline and filtering out the noise and taking what's useful. This is same as our society and daily social interactions.

3.2 User Research

The idea here is pretty straightforward: we want to figure out the best way for AI to console people who are going through something emotionally heavy. The plan is to split participants into two groups and have them talk to the AI about a deep or charged topic. One group gets the "reflective questioning" treatment, where every third response the AI throws in something like "What is the strongest counter-argument to your current belief?" The whole point is to push people toward actually thinking through their position instead of just venting into the void. The other group gets the opposite approach, where the AI affirms their feelings and gently steers the conversation toward a more positive or constructive light. At the end of the session, we give everyone a questionnaire asking how they feel mentally about the topic, whether the interaction helped their mindset, and how they now perceive the opposing point of view. By comparing these two completely different approaches, we can start to see which strategy actually works better for consoling people. But it goes deeper than that too. Since the questioning style and the leading style are opposite ends of the spectrum, we can also experiment with the timing and frequency of the reflective questions, like does asking after every third response feel overwhelming, or would every fifth be better? Maybe some people respond well to being challenged early on but need affirmation later. The data from both groups would let us figure out the right balance, the right moment to interject, and how people actually feel after being asked to consider the other side versus being guided toward positivity. After the whole session ends, we can also check whether engaging with the opposite point of view actually helped their emotional state or just made things more frustrating. That comparison is the real goal here: finding the most effective,

least harmful way for an AI to support someone through a difficult conversation.

To test this, I'd recruit frequent, AI-savvy users (people who actively interact across diverse topics like social sciences, arts, sciences, travel, and finance). As this group is most likely to understand the AI's limitations, exposing them to potentially wrong interpretations carries far less risk than experimenting this with non-savvy users who might take everything too seriously. Pariticpant bias is unavaoidable. The group is inherently AI-savvy and frequent users, so the results might not apply to casual or first-time users. There's also self-selection, where people comfortable discussing emotions will sign up, which skews toward the mentally resilient. And participants might change their natural behavior simply because they know they're being watched.

Lesson learnt from Facebook experiment is to clearly state the experiment is being carried out and being transparent about it. The recruitment message would go something like this: "Let's help the world! We are testing a new 'Reflective Dialogue' feature. If you are 18+ and have used our platform for personal advice in the last 30 days, join a 3-session study (15 min total). You will receive 10 free Deep Research queries as a thank you. Click here to read the consent form and enroll." Before any discussion begins, a consent message would pop up: "You will be randomly assigned to either a standard-response group or a 'Reflection' group (you will not know which). Your anonymized conversation text will be analyzed to measure emotional clarity. Risks include mild discomfort when reflecting. No deception is used; you will be fully debriefed after your third session. You may withdraw at any time by closing the chat. By clicking 'I Consent,' you agree to participate voluntarily."

4 QUESTION 4

View the video: *https://mediaspace.gatech.edu/media/t/1_nv5lf0e9*